

Coping with segmental–prosodic incongruity in spoken word recognition in Japanese

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Abstract

A spoken word consists of two phonological cues: segments (phonemes) and prosody (e.g., lexical pitch accent). While both segments and prosody contribute to spoken word recognition, it is unknown how listeners interpret a spoken word when these cues contradict each other. For example, a word **tansu* HLL is logically interpreted as both a segmental mispronunciation of *dansu* HLL “dance” and a prosodic mispronunciation of *tansu* LHH “wardrobe.” The present study investigated how Japanese listeners interpret those segmentally and prosodically incongruent words in spoken word recognition using a cross-modal repetition priming paradigm. When ISI = 0 ms, listeners’ lexical decision was delayed more for prosodic mispronunciation (prime: **tansu* HLL; target: *tansu* LHH “wardrobe”) than for segmental mispronunciation (prime: **dansu* LHH; target: *tansu* LHH). In contrast, when ISI = 750 ms, listeners’ lexical decision was delayed more for segmental mispronunciation than for prosodic mispronunciation. These results demonstrate that lexical access to segmentally and prosodically incongruent words is a dynamic process, in which listeners first try a prosody-oriented interpretation and then shift to a segment-oriented interpretation.

Index Terms: spoken word recognition, lexical prosody, pitch accent, Japanese

1. Introduction

Spoken words consist of multiple phonological cues. For example, a Japanese word *tansu* “wardrobe” is produced with a serial combination of five phonemes, namely, /t/, /a/, /n/, /s/, and /u/. Simultaneously, it is produced with an unaccented pitch accent pattern (LHH; L or H indicates a high or low tonal pattern for each mora). In contrast, the word *dansu* “dance” is produced with a segmental combination of /d/, /a/, /n/, /s/, and /u/ and simultaneously with an accented pitch accent pattern (HLL). Both segments and lexical prosody are critical for lexical distinction. How do listeners interpret words when these two phonological cues contradict one another? In other words, when they listen to a mispronounced word **tansu* HLL, do they interpret it as *tansu* LHH “wardrobe” or *dansu* HLL “dance”?

The role of prosodic information, especially lexical prosody, in spoken word recognition has been debated in the literature of psycholinguistics [1–3]. Empirical studies have shown that lexical stress plays a minor role for lexical access in English [4–6]. In contrast, studies in Dutch [7, 8], Spanish [9], Chinese [10, 11], and Japanese [12–15], have demonstrated that lexical prosodic cues are important for lexical access in the spoken word recognition mechanisms of these languages.

A crucial role of lexical prosody in spoken word

recognition is discriminating words that are distinguished solely by lexical prosodic information (i.e., minimal pairs such as *jidoo* HLL “children” and *jidoo* LHH “automatic” [12, 15]) during lexical access. In addition to the independent use of lexical prosody, it also influences segmental identification. Connine and others [16] demonstrated that when a /t–d/ ambiguous sound was presented in the context of *_igress*, the perceptual categorization of the ambiguous segment was biased toward being voiced (/d/) if the stress pattern was weak–strong. In contrast, categorization was biased toward being voiceless (/t/) if the stress pattern was strong–weak. This biasing effect occurred because both *diGRESS* and *Tlgress* (capital letters indicate stressed syllables) are real words whereas **tiGRESS* and **DiGRESS* are nonwords.

Similar effects are observed in Japanese, where pairs contrasted by both segments and lexical pitch accent, such as *dansu* HLL “dance” and *tansu* LHH “wardrobe,” are used [17]. These effects suggest that when segmental input is acoustically ambiguous, lexical prosodic information aids in segmental perception so that the entire word would become a real word.

A question then remains as to whether the effect of lexical prosody on segmental perception occurs only when segmental perception itself suffers from the acoustic ambiguity of the sound. Given the redundancy of acoustic cues in speech recognition, it is reasonable to assume that lexical prosody compensates for segmental processing. However, because segments and lexical prosody are independent of each other in production, we can systematically produce equivocal mispronunciations such as **tansu* HLL. It remains unclear whether listeners rely on segmental information or lexical prosodic information to interpret these ambiguous words.

With respect to spoken word recognition based on mispronounced word input, Lee [11], using a cross-modal priming paradigm, suggested that mispronunciation of both segments and tones prevents lexical access in Mandarin Chinese. However, segmental mispronunciation with tonal overlap (e.g., prime: *cang2* “hide”; target: *lou2* “hall”) induced a greater increase in response time than tonal mispronunciation with segmental overlap (e.g., prime: *lou3* “hug”; target: *lou2* “hall”). This suggests qualitative differences between the mispronunciation of segments and that of prosody in the online processing of spoken word recognition.

However, since the importance of lexical prosody in spoken word recognition differs across languages [18–20], how spoken word recognition is influenced by segments and prosody, as well as by their mispronunciation, may be different in Japanese. In addition, Lee’s study [11] tested mispronunciation that is identifiable as segmental or prosodic. With respect to the equivocal mispronunciation mentioned above, how listeners interpret words based on incongruous segmental and prosodic

information is unknown.

To address these questions, we investigated the process of lexical access when listening to segmentally and prosodically incongruent words, such as **tansu* HLL. We used a repetition priming paradigm, which is an established method for assessing the process of lexical access [7, 9, 11, 13, 15]. This paradigm enables us to evaluate the dynamic process of lexical access by adjusting prime–target timing (i.e., inter-stimulus interval: ISI) [5, 21, 22]. We conducted two experiments using different ISIs.

2. Experiment 1

2.1. Methods

2.1.1. Participants

Thirty-two native speakers of Tokyo Japanese (18–53 years old; mean = 25.12; $SD = 6.74$) participated in the experiment. They self-identified as being from the Kanto area in Japan. They received monetary compensation for their participation.

2.1.2. Materials

Twelve minimal pairs that contrasted both the voicing of the initial consonant (voiced or voiceless) and the lexical pitch accent pattern (accented or unaccented), e.g., *dansu* HLL “dance” and *tansu* LHH “wardrobe,” were prepared. These pairs are the same as those used in a previous study [23]. All of these words were highly familiar, and the familiarity did not differ across pairs [24].

The item words were chosen so that the voicing of the initial consonant was lexically correlated with the pitch accent pattern. Thus, alternating either the voicing of the initial consonant or the lexical pitch accent results in ambiguous mispronunciation. For example, **tansu* HLL is regarded as both a segmental mispronunciation of *dansu* HLL “dance” and a prosodic mispronunciation of *tansu* LHH “wardrobe.” Similarly, **dansu* LHH is regarded as both a segmental mispronunciation of *tansu* LHH and a prosodic mispronunciation of *dansu* HLL.

To assess the process of lexical access when listeners listened to ambiguous mispronounced words, we set four conditions within the cross-modal repetition priming paradigm, as shown in Table 1. The congruent condition contained a correctly pronounced word as an auditory prime and an identical word as a visual target (*tansu* LHH – *tansu* LHH). In the prosodically mispronounced condition, the pitch accent pattern of the auditory prime was replaced with the other, and the visual target remained the same (**tansu* HLL – *tansu* LHH). In the segmentally mispronounced condition, the voicing of the initial consonant was replaced with the other (**dansu* LHH – *tansu* LHH). Finally, the baseline condition contained a phonetically and semantically unrelated word as an auditory prime (*sabaku* “desert” – *tansu* LHH).

Table 1: An example of experimental materials.

Condition	Prime	Target
congruent	<i>tansu</i> LHH	
prosodically mispronounced	<i>tansu</i> HLL	箆笥
segmentally mispronounced	<i>dansu</i> LHH	<i>tansu</i> “wardrobe”
baseline	<i>sabaku</i>	

The audio files were recorded by an experienced female native Tokyo Japanese speaker, who is familiar with Japanese phonetics, in a sound attenuated room. The production was digitized at a sampling rate of 44.1 kHz and with 16-bit resolution in Praat 6.3.19 [25] with a Samson Q2U microphone. Prime words were originally produced with a carrier phrase *sore wa* “it is ____.” The carrier phrase was removed afterwards. Words with mispronunciation were recorded in the same manner. Another native Tokyo Japanese listener familiar with Japanese phonetics confirmed by auditory impression that the mispronounced words were *correctly* pronounced with the other pitch accent pattern in the prosodically mispronounced condition, and that they were *correctly* pronounced with the other voicing of the initial consonant in the segmentally mispronounced condition. The intensities of all materials were normalized.

The experimental items included unaccented items (*tansu* LHH in the congruent condition) and accented items (*dansu* HLL in the congruent condition) of the pairs. The critical trials were 24 in total. However, since we aimed to obtain more reliable effect sizes with small sample sizes, and most previous studies did not report that the items’ pitch accent pattern affected response [12, 15], we did not include this pitch accent pattern as a factor in the experiment. To avoid habituation caused by repeating trials with similar words, the participants experienced one of four conditions per item. The conditions were counterbalanced using a Latin square design. We also added 72 filler trials, 48 of which had nonword targets (nonword proportion was 50%).

2.1.3. Procedure

The experiment was conducted online in PClbex [26]. The participants used their own computer and listened to the auditory materials at their comfortable level through headphones or earphones. The experiment lasted for 30 min.

The main task was a lexical decision task in the cross-modal priming paradigm, accompanied by a dummy auditory comprehension task. The participants first completed 6 practice trials and then performed 96 test trials with an optional break.

At the beginning of each trial, a fixation point (+) was presented at the center of the display. The auditory prime was played after 1000 ms. The participants were instructed to listen to it carefully. At the offset of the auditory prime, the fixation point disappeared, and the visual target was subsequently presented. The inter-stimulus interval (ISI) between primes and targets was thus 0 ms. Target words were presented in their common form, i.e., kanji. However, foreign words (e.g., *dansu* “dance”) were presented in *katakana* (Japanese syllabary characters). The participants were asked to judge whether the target words were real words or nonwords and press F for word response and J for nonword response. The participants were asked to press the key as rapidly and accurately as possible. The timeout for the lexical decision was 3000 ms. After the lexical decision, participants answered whether the auditory prime word they listened to during the trial was trimoraic or tetramoraic (some filler items contained tetramoraic prime words). This was a dummy task to confirm the listeners’ attention to the prime. The dummy task was a forced-choice task without timeout. After the response, the next trial started automatically. The interval between trials was 1000 ms.

After the lexical decision task, a follow-up production task was conducted to confirm participants’ knowledge of the pitch accent patterns of the experimental items. In this task,

participants pronounced all the items used as prime words based on their intuition.

2.2. Results and Discussion

The responses in the follow-up production task were regarded as correct if the participants produced words appropriately in the pitch accent pattern of Tokyo Japanese. Based on the results of the production and lexical decision tasks, data from three participants were not used for the reaction time analysis because of their low response accuracy (less than 80%). In the reaction time analysis, reaction times less than 300 ms or more than 1600 ms were regarded as outliers and removed from the data. Additionally, individual data points from trials in which the participants made incorrect responses in the lexical decision task or containing primes that the participants mispronounced in the production task were excluded.

The reaction times were analyzed by a linear mixed-effects (LME) model in R 4.4.0 [27]. The fixed effect was the prime condition. Comparisons between the congruent and baseline conditions, between the prosodically mispronounced and baseline conditions, and between the segmentally mispronounced and baseline conditions were coded as 0.75 and -0.25, respectively. The slopes and intercepts of both participants and items were included as random effects. An optimal model was determined by backward selection using the *step* function. A maximal model containing all the effects was initially calculated and the random effects of least variance were iteratively removed. After choosing the optimal model, data points with an absolute residual greater than 2.5 were additionally removed from the final model.

The mean reaction times for each condition are shown in Figure 1. The LME model (formula: $RT \sim \text{Prime Pattern} + (1 | \text{participant})$) suggested that the mean reaction time for the congruent condition was significantly shorter than that for the baseline condition ($\beta = -83.207$, $SE = 27.696$, $df = 484.781$, $t = -3.004$, $p = .003$). This decrease in reaction time reflected a repetition priming effect, where response to target words was facilitated by preceding presentation of the prime words. However, the mean reaction time for the prosodically mispronounced condition was significantly longer than that for the baseline condition ($\beta = 61.533$, $SE = 27.328$, $df = 477.022$, $t = 2.252$, $p = .025$), whereas there was no significant difference between the mean reaction times for the segmentally mispronounced condition and the baseline condition ($\beta = 48.430$, $SE = 28.744$, $df = 481.493$, $t = 1.685$, $p = .093$).

The reaction times for the prosodically mispronounced condition and the segmentally mispronounced condition indicated that mispronunciation of lexical prosody or segments disturbed appropriate lexical access to intended words, as evidenced by the lack of priming effects in these conditions. While both segmental and prosodic mispronunciation affected lexical access, the response to segmental mispronunciation was neutral compared with the response to the unrelated prime (i.e., the baseline condition). In contrast, the response to prosodic mispronunciation resulted in greater inhibition.

This difference is interpreted as an indication that when segmental-prosodic incongruity exists in the auditory words, listeners tend to interpret them based on prosodic rather than segmental information. The use of prosodic information to resolve incongruity resulted in greater inhibition in the prosodically mispronounced condition (**tansu* HLL) and relatively less inhibition in the segmentally mispronounced condition (**dansu* LHH).

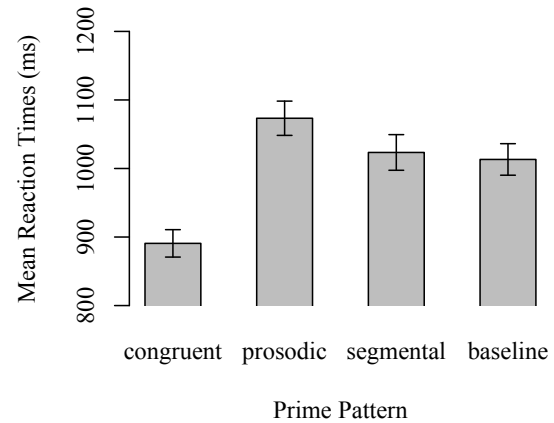


Figure 1: Mean reaction times for each condition in Experiment 1 (ISI = 0 ms). Error bars indicate standard errors of the mean. “Prosodic” refers to the prosodically mispronounced condition and “segmental” refers to the segmentally mispronounced condition.

However, since spoken word recognition is a dynamic process, in which the cues used for lexical competition change over time [28, 29], it is plausible that lexical prosody is utilized only at the offset of word presentation (ISI = 0 ms) and that its priority usage may be weakened at another time of lexical access. To address this possibility, we used delayed priming at an ISI of 750 ms in Experiment 2. In previous studies, a 750 ms delay allowed the evaluation of the postlexical process of spoken word recognition, in which contextual information overrides contextually incongruent candidates [5, 22]. In a study in Japanese, listeners reinterpreted mispronounced lexical prosody within this latency [21]. In Experiment 2, we changed the ISI of the target presentation from 0 ms to 750 ms after the offset of the prime word.

3. Experiment 2

3.1. Methods

3.1.1. Participants

Thirty-four native speakers of Tokyo Japanese (19–41 years old; mean = 26.12; $SD = 5.79$) participated in the experiment. They self-identified as being from the Kanto area in Japan. They received monetary compensation for their participation.

3.1.2. Materials

Prime and target items used in Experiment 2 were identical to those used in Experiment 1.

3.1.3. Procedure

The procedure was the same as that used in Experiment 1, except that the ISI between primes and targets was set to 750 ms to evaluate the postlexical process of spoken word recognition when listening to mispronounced materials.

3.2. Results and Discussion

Data from seven participants were not used in the analysis because of their low response accuracy (less than 80%) in the production or lexical decision tasks. Outliers were excluded using the same criteria as in Experiment 1. Coding and model

selection in the LME analysis were performed in the same manner as in Experiment 1.

The mean reaction times for each condition are shown in Figure 2. The LME model (formula: $RT \sim \text{Prime Pattern} + (1 | \text{participant}) + (1 | \text{item})$) suggested that the mean reaction time for the congruent condition was marginally significantly shorter than that for the baseline condition ($\beta = -50.662$, $SE = 26.623$, $df = 422.068$, $t = -1.903$, $p = .058$). This facilitation, not conclusively, indicated that the repetition priming effect remained 750 ms after the offset of the prime words. In contrast, the mean reaction time for the segmentally mispronounced condition was significantly longer than that for the baseline condition ($\beta = 132.132$, $SE = 28.865$, $df = 422.153$, $t = 4.578$, $p < .001$), whereas there was no significant difference between the mean reaction times for the prosodically mispronounced condition and for baseline condition ($\beta = 17.300$, $SE = 27.277$, $df = 421.946$, $t = 0.634$, $p = .526$).

We conducted an additional analysis comparing the reaction times for each condition between Experiment 1 and 2. The prime conditions, experiments, and their interaction were set as fixed effects. Comparison of the experiments was coded as -0.5 (Experiment 1) and 0.5 (Experiment 2). The optimal LME model (formula: $RT \sim \text{Prime Pattern} \times \text{Experiment} + (1 | \text{participant})$) suggested that the simple main effects of the congruent condition ($\beta = -67.657$, $SE = 19.266$, $df = 906.736$, $t = -3.512$, $p < .001$), the prosodically mispronounced condition ($\beta = 40.605$, $SE = 19.336$, $df = 898.995$, $t = 2.100$, $p = .036$), the segmentally mispronounced condition ($\beta = 87.929$, $SE = 20.398$, $df = 903.367$, $t = 4.311$, $p < .001$), and the interaction between the segmentally mispronounced condition and the experiment ($\beta = 82.829$, $SE = 40.852$, $df = 902.983$, $t = 2.028$, $p = .043$) were significant. However, the main effect of experiments and the other interactions were not significant (Experiment: $\beta = 34.326$, $SE = 45.884$, $df = 53.617$, $t = 0.748$, $p = .458$; congruent $\times$ Experiment: $\beta = 31.965$, $SE = 38.489$, $df = 906.018$, $t = 0.831$, $p = .406$; prosodically mispronounced $\times$ Experiment: $\beta = -44.452$, $SE = 38.712$, $df = 899.020$, $t = -1.148$, $p = .251$).

The inhibition evoked by mispronunciation appeared in the opposite direction to that in Experiment 1. When the ISI was 750 ms, the response to segmental mispronunciation resulted in

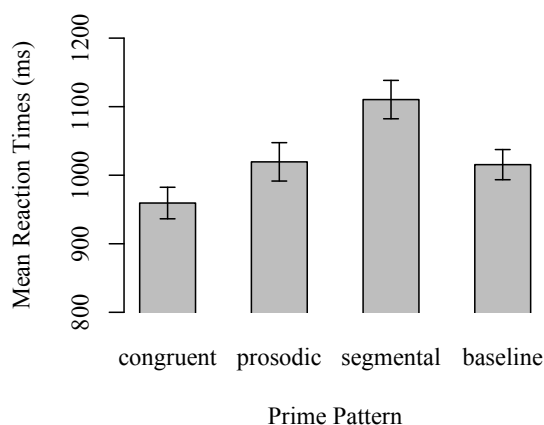


Figure 2: Mean reaction times for each condition in Experiment 2 (ISI = 750 ms). Error bars indicate standard errors of the mean. “Prosodic” refers to the prosodically mispronounced condition and “segmental” refers to the segmentally mispronounced condition.

greater inhibition than that to prosodic mispronunciation. This finding indicates that an attempt to interpret words based on segments *ignoring* prosodic information was made in the postlexical process 750 ms after the offset of the prime presentation. The interaction effect between the segmentally mispronounced condition and experiment also suggests that a segment-based interpretation emerged after the initial process of spoken word recognition. These results are in line with those of a previous study in which mispronunciation of lexical prosody was resolved within 750 ms with reference to contextual information [21]. The present results further suggest that recognition *without* lexical prosody occurs during the postlexical process when there is segmental–prosodic incongruity in the presented words.

4. General Discussion

Two cross-modal priming experiments demonstrated that participants’ responses were inhibited by prosodic mispronunciation at ISI = 0 ms but by segmental mispronunciation at ISI = 750 ms. These results suggest that lexical access to segmentally–prosodically incongruent words is a dynamic process in which listeners first attempt a prosody-oriented interpretation and then shift to a segment-oriented interpretation within 750 ms. When two phonological cues, segments and lexical prosody, apparently contradict each other, although listeners try both interpretations based on one of these cues, the cue-weighting for each piece of information changes over time.

The priority usage of lexical prosody in equivocal situations is in line with the observation that it constrains the initial stage of lexical access [7–10, 12, 15]. The present study further demonstrates that lexical prosody overrides segmental mispronunciation during the initial lexical access. This may be because lexical prosody is lexical information, whereas individual segments are not. Since lexical prosody provides more reliable information for identifying the spoken words, listeners heuristically rely more on lexical prosodic information than on segmental information when these two cues are incongruent. However, cue-weighting is completely reversed in later stages. Since it is logically possible that lexical prosody is mispronounced, listeners interpret words regardless of lexical prosodic information in the postlexical process 750 ms after the offset of the prime presentation.

In the present study, however, the 750 ms latency to assess the postlexical process in Experiment 2, which was derived from English studies [5, 22], was arbitrary. Validation of this latency in Japanese spoken word recognition and identification of the precise timing for reanalysis must be addressed in future studies.

Even with these limitations, the present study potentially challenges traditional spoken word recognition models that do not utilize prosodic information [30–32], suggesting that lexical prosodic information is important for spoken word recognition [3], particularly in Japanese. Moreover, the present study supports the dynamic process of lexical competition [28, 29] against segmental and lexical prosodic cues, showing both rigidity and flexibility to lexical prosodic information that changes over the time course of spoken word recognition.

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6. References

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